

Clovis SFEIR

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Profile

IMT Atlantique engineering student with a quantitative-reasoning and competitive-problem-solving background. Placed **24th out of 124 teams** (top 20%) in the HRT/Partcl Macro Placement Challenge 2026, a live optimization competition against Stanford, Princeton, and ETH Zürich entrants. Built five systems (C++20/Rust/Python) benchmarked against industry references or labeled ground-truth, including a commodity spread-trading model built on mean-reversion and statistical signal validation.

Education

IMT Atlantique – Engineering Degree in Software Engineering (Apprenticeship) *Sep 2025 – Jun 2028*
Nantes, France *Algorithms & Data Structures, OS, Computer Networks, Distributed Systems, Cloud Computing, AI*
Bachelor in Computer Science *Sep 2022 – Jun 2025*
La Rochelle, France *Class rank: 1/85 | Software Engineering, Databases, Cybersecurity*
HRT Macro Placement Challenge 2026 (Hudson River Trading/Partcl) — **24th/124** (top 20%); hybrid analytical placer: force-directed global placement with timing-aware legalization; proxy cost **1.095** beats SA baseline (2.125) and RePIAce (1.458); field includes Stanford, Princeton, ETH Zürich.

Projects

Crude Oil Spread Trading Engine *C++20, stochastic processes, order book*

- Models refining margins (3:2:1 crack spread, generalized N:M:K) and mean-reverting inter-crude spreads via an Ornstein-Uhlenbeck process (Euler-Maruyama) with a rolling z-score signal – real commodity-market economics, not a toy simulation.
- OU stationary distribution verified against theory (500K-step sim); crack-spread formula checked to 10^{-9} ; matched by a price-time-priority order book; 58 assertions/21 tests, ASan+UBSan clean.

IIoT Predictive Maintenance *Rust, tokio, EWMA/CUSUM*

- Async telemetry pipeline (**tokio** mpsec) fuses EWMA + CUSUM control charts across 3 sensor channels to flag developing trend shifts early – reading noisy signals and deciding when a shift is real, not noise.
- Sensor-fusion tuning (2-of-3 channels + widened control limits) cut false positives from a 0.41–0.50 precision baseline to **0/10 across 10 seeds**; precision **1.0000**, recall 0.9642, F1 0.9818; 26 tests, 0 failures.

Engineering Docs RAG *Python, BM25, information retrieval*

- From-scratch Okapi BM25 retrieval engine (chunking + ranking) built to accelerate search over dense technical/procedural documents; pluggable extractive/LLM generator interface.
- Hand-derived BM25 scoring verified to 9 significant figures; measured **precision@1 0.90**, recall@5 1.00, **MRR 0.944** on a 40-question labeled set; 53 tests, zero runtime dependencies.

HFT Limit Order Book Matching Engine *C++20, cache-aware layout, intrusive data structures*

- Price-time-priority matching engine (limit/market/cancel) with intrusive doubly-linked price-level lists (**O(1) cancel**) and pre-allocated object pooling – built to understand the mechanics behind every quote and fill on a trading screen.
- Benchmark suite across three synthetic order-flow regimes: **2.3M+ msg/s** single-threaded throughput, **P50 0.6–1.0 μ s** latency, zero hot-path allocations; unit tested.

ICS Intrusion Detection (IEC 60870-5-104) *C++20, protocol parsing, explainable rules*

- Rule-based detector for IEC 60870-5-104 mapped to real attack patterns: control commands to unexpected breaker objects, sequence-number gaps, u-format/command floods.
- Parses APCI+ASDU at **39.5 ns (25.3 M msg/s)**; **100% recall, 0 false positives** on a labeled synthetic trace; 153 assertions/37 tests, ASan+UBSan clean.

Work Experience

Infotel *Nantes, France*
AI Engineer Apprentice – Lab IA *Sep 2025 – Jun 2028*

- Built a Rust-based autonomous software-delivery harness that takes a complete specification through architecture synthesis (subsystems, interface contracts), epic/slice decomposition, and TDD-first generation to a tested, deployable application end-to-end; automated compile+test quality gates with iterative self-repair, independent code-review pass, and cross-slice conflict detection before delivery; 928 Rust tests.

Excelia Group *La Rochelle, France*
Software Developer Intern / AI Software Engineer Intern *Two internships: Jan–Mar 2025 & May–Jul 2024*

- RAG pipeline (chunking, OpenAI embeddings, flat FAISS, cross-encoder re-ranking): **40% reduction** in lookup time vs. keyword-search; cut p95 latency 30–40% via embedding cache + asyncio parallelisation; Prometheus/Grafana, Docker, GCP.
- Real-time interview simulator (Next.js/React, GPT-4o, ElevenLabs TTS): streaming responses, scenario-branching state machine, A/B prompt-scoring harness.

Technical Skills

ML / Quant	Ornstein-Uhlenbeck, EWMA/CUSUM change-detection, Kalman filter, BM25/information retrieval, LightGBM/XGB/CatBoost
Languages	Python, C++20, Rust, TypeScript, SQL
Perf / Systems	cache-aware layout, intrusive linked lists, acquire/release atomics, ICS/SCADA protocol parsing
Infrastructure	Docker, tokio async, Prometheus/Grafana, GitHub Actions CI, GCP, Redis
Frameworks	Flask, Node.js/Express, React/Next.js, REST/gRPC